AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 22 · STUDENT LESSON

Nearest Neighbor Reasoning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Measure the gaps. Select the neighbours. Vote before scoring.

Use stored examples under declared k and tie rules without leaking query answers into prediction.

Keep one habit: Training pool → numeric ordering → k comparison → frozen validation and baseline.

MAIN LESSON · SECTION 1

Which stored examples vote, and under which tie rule?

> Order by distance before selecting neighbours or looking at query references.

Your learning goals

I can compute absolute **distance**s and rank **training neighbour**s.

I can apply a declared **neighbour** count and **distance tie rule**.

I can evaluate **neighbour prediction**s without using validation labels as **neighbour**s.

Recall a stored tree before comparing methods

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What does a fitted depth-one tree store, and what does it do with a new example?

Your end goal

By the end, I can order **training neighbour**s, apply k and keep query labels out of **prediction**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 22: Nearest-**Neighbor** Reasoning

A new item can be predicted from nearby labeled examples.
</details>



THINKER MISSION

Run a paper k-nearest-**neighbor**s **vote** and resolve a tie.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Store labelled training examples and declare the method

> IDs identify rows and break distance ties; they are not measured features.

The nearest-neighbour prediction idea

A nearest-**neighbour** classifier predicts using nearby **training** examples rather than following a fitted tree.

This week we store labelled examples, measure how different their feature values are from a new query, and let selected **neighbour**s **vote**.

Nearby means close under our chosen feature and **distance**, not necessarily similar in every real-world way.

The complete five-row training pool

Our new fictional dataset has a **distance** reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity.

IDs identify rows and break **distance** ties; they are not numeric features. These invented readings are not a robot safety system.

ID	Reading in cm	Label
A	2	blocked
B	4	clear
C	8	clear

ID	Reading in cm	Label
D	10	blocked
E	12	clear

Distance, k and the predefined ID rule

For this one-dimensional feature, **distance** is abs(**training** reading minus query reading). abs removes the sign. Set k=3 and give each selected **neighbour** one **vote**.

Order equal **distance**s by ID alphabetically, declared before scoring. In this binary, three-**vote** exercise the label **vote** cannot tie, but the **distance**s can.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use nearest-**neighbor** reasoning to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace sorting, selection and voting in complete Python

> Keep the labels out of distance calculation.

Predict and execute the single-query script

```
train = [("A", 2, "blocked"), ("B", 4, "clear"),
         ("C", 8, "clear"), ("D", 10, "blocked"),
         ("E", 12, "clear")]
query = 3
ranked = sorted((abs(x - query), item_id, label)
                for item_id, x, label in train)
near = ranked[:3]
labels = [row[2] for row in near]
prediction = "clear"
if labels.count("blocked") > labels.count("clear"):
    prediction = "blocked"
```

```
print([row[1] for row in near])
print(prediction)
```

Read tuple sorting and the first-three slice

For query 3, **distance**s to A through E are 1,1,5,7,9 cm. sorted orders tuples by **distance** first, then the unique ID. ranked[:3] takes the first three rows.

The **output** is ['A', 'B', 'C'] then clear: one blocked **vote** and two clear **vote**s.

Explain what changes for k=1

If k were 1 under the same **distance tie rule**, A would win and predict blocked. Changing k can change a **prediction**.

We declared k=3 for this exercise; do not choose k after seeing the desired answer and pretend the method was unchanged.

Check the script's declared assumptions

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three **training** rows. It does not validate those conditions.

A larger k than the stored row count would not produce the promised number of **vote**s. Keep unresolved review labels under a separate documented policy.

Point to the deciding step

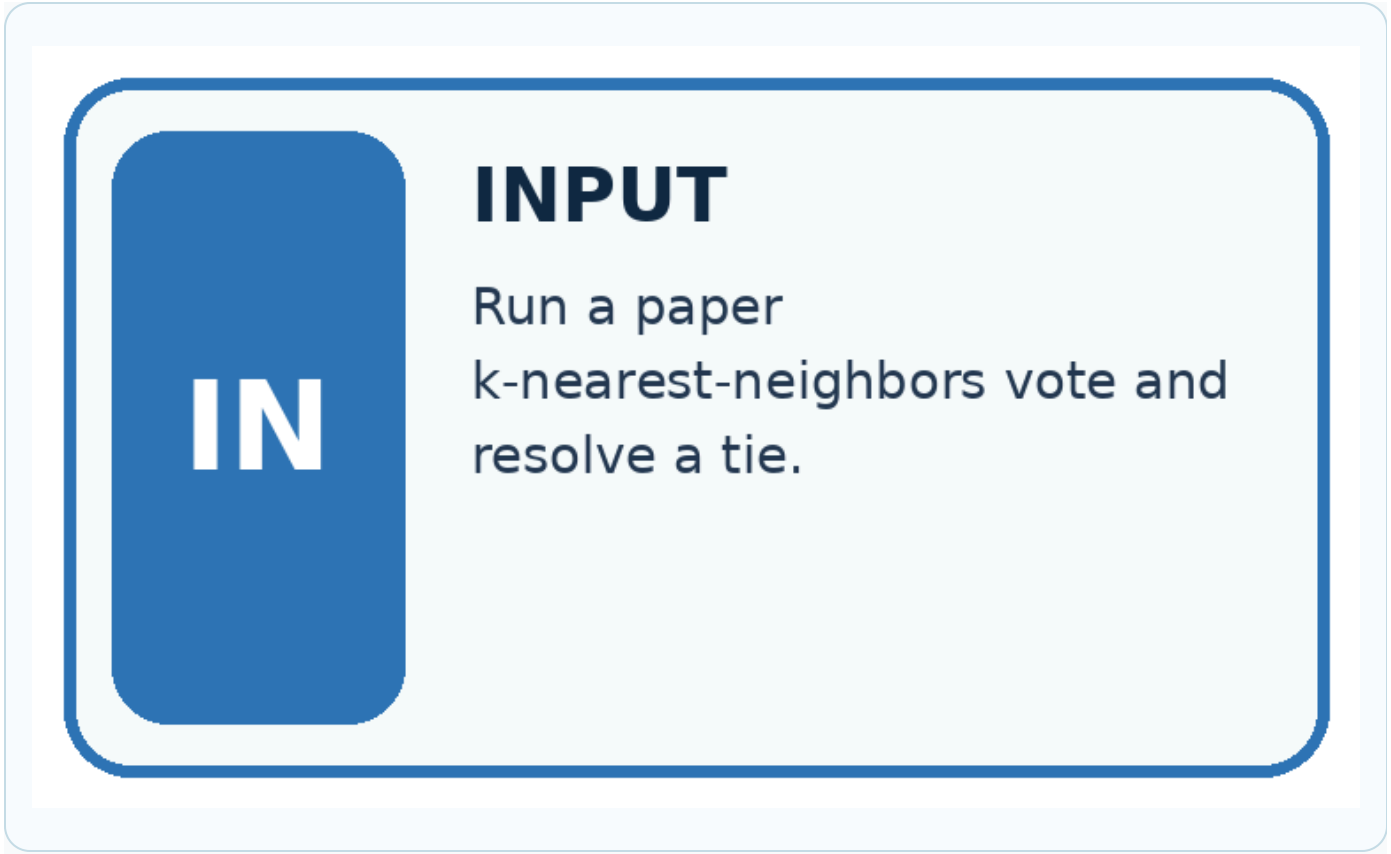
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

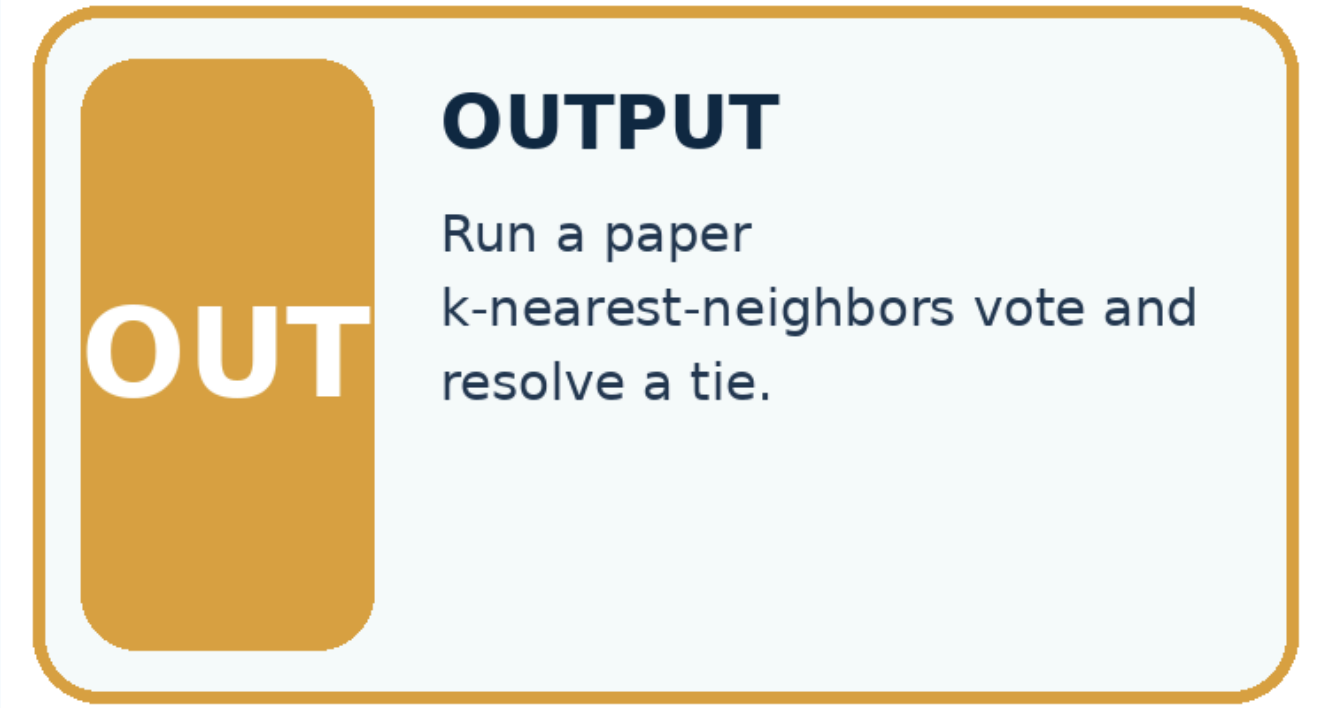
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Run a paper k-nearest-neighbors vote and resolve a tie.
↓
PROCESS
A new item can be predicted from nearby labeled examples.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

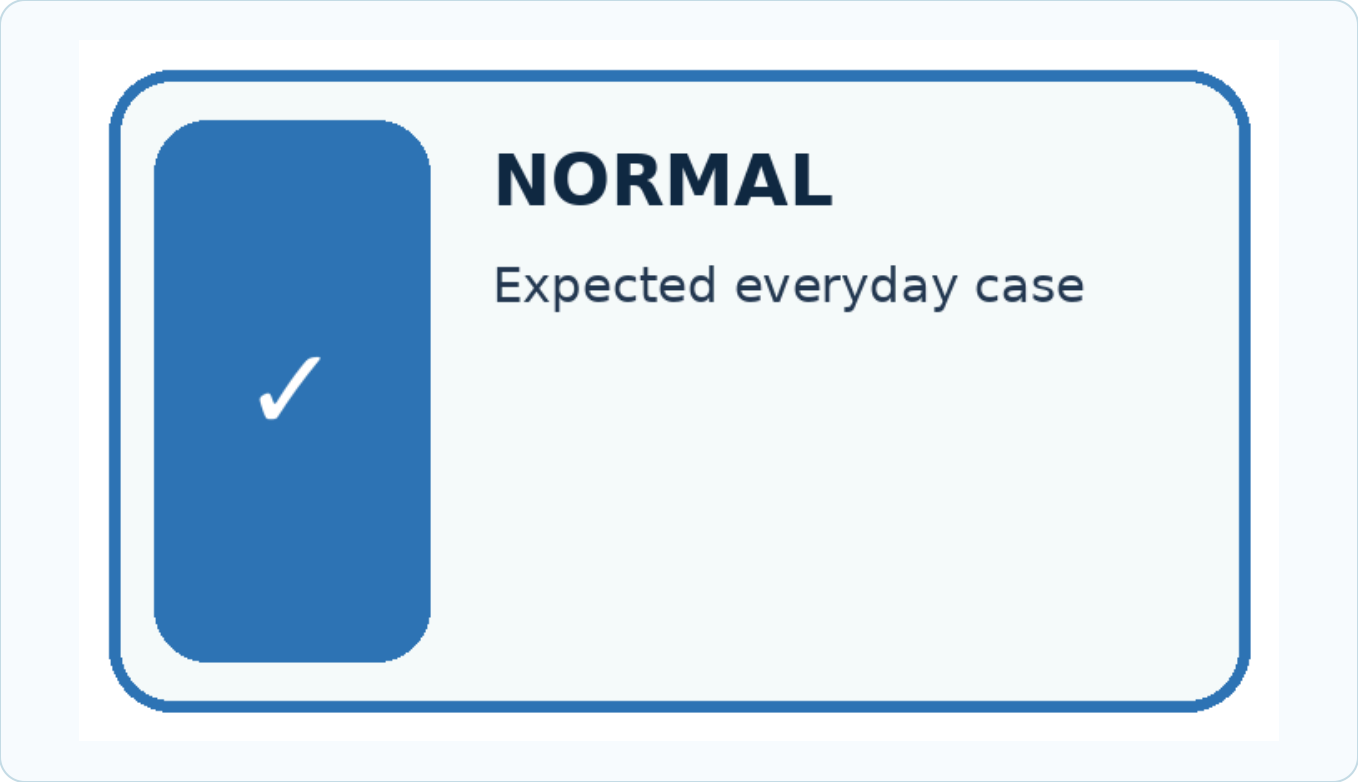
A closer explanation

Connect the picture and the explanation

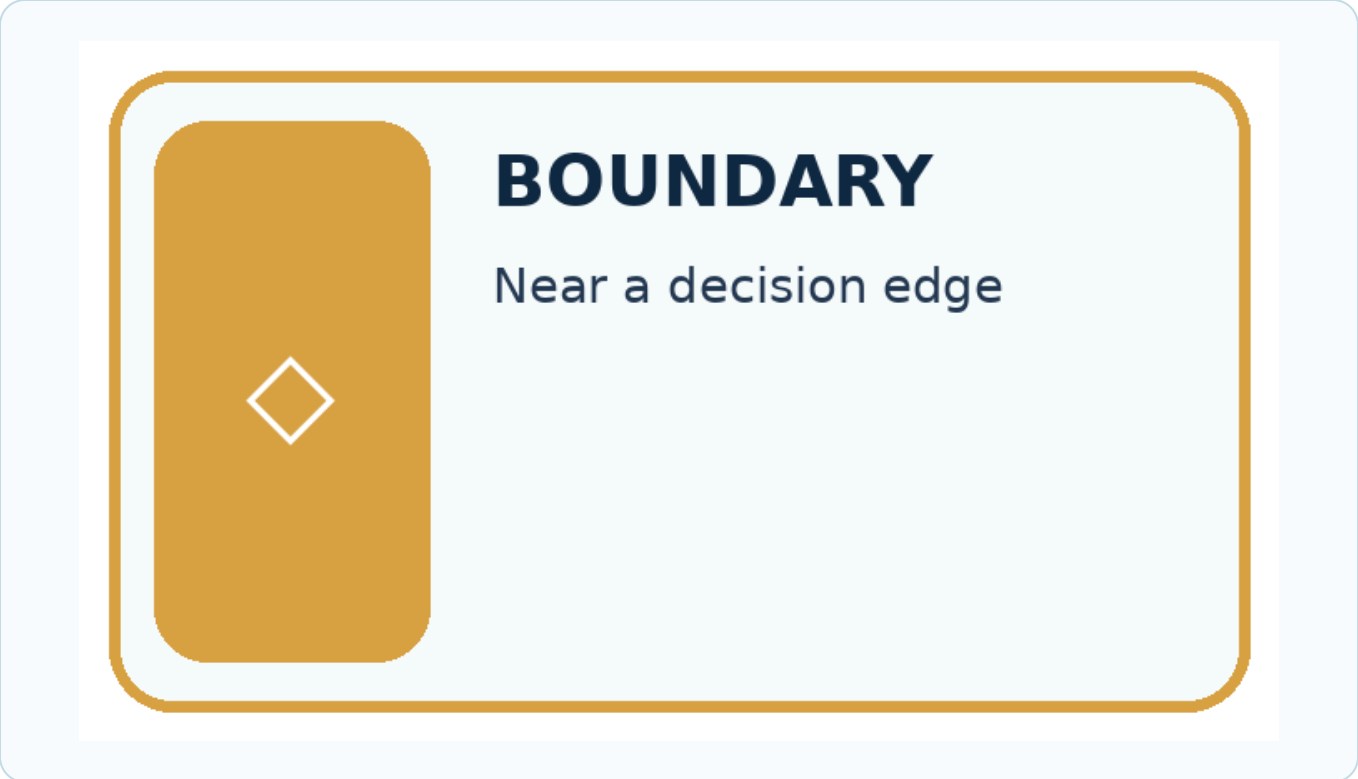
Compare a distance tie with a vote

> A reproducible tie policy is not evidence that the chosen label is true.

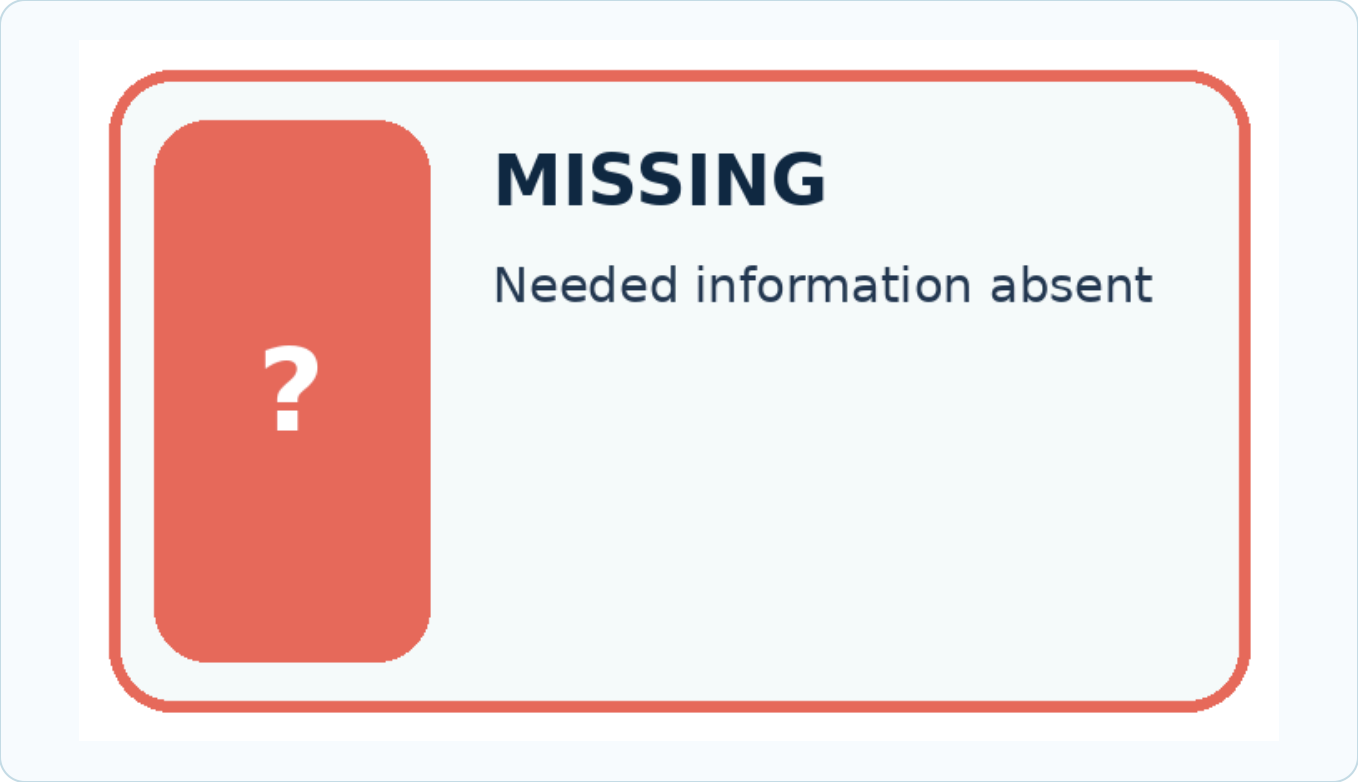
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Move query from 7 to 9 while keeping **training** fixed.

▼ Check after predicting

Prediction: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

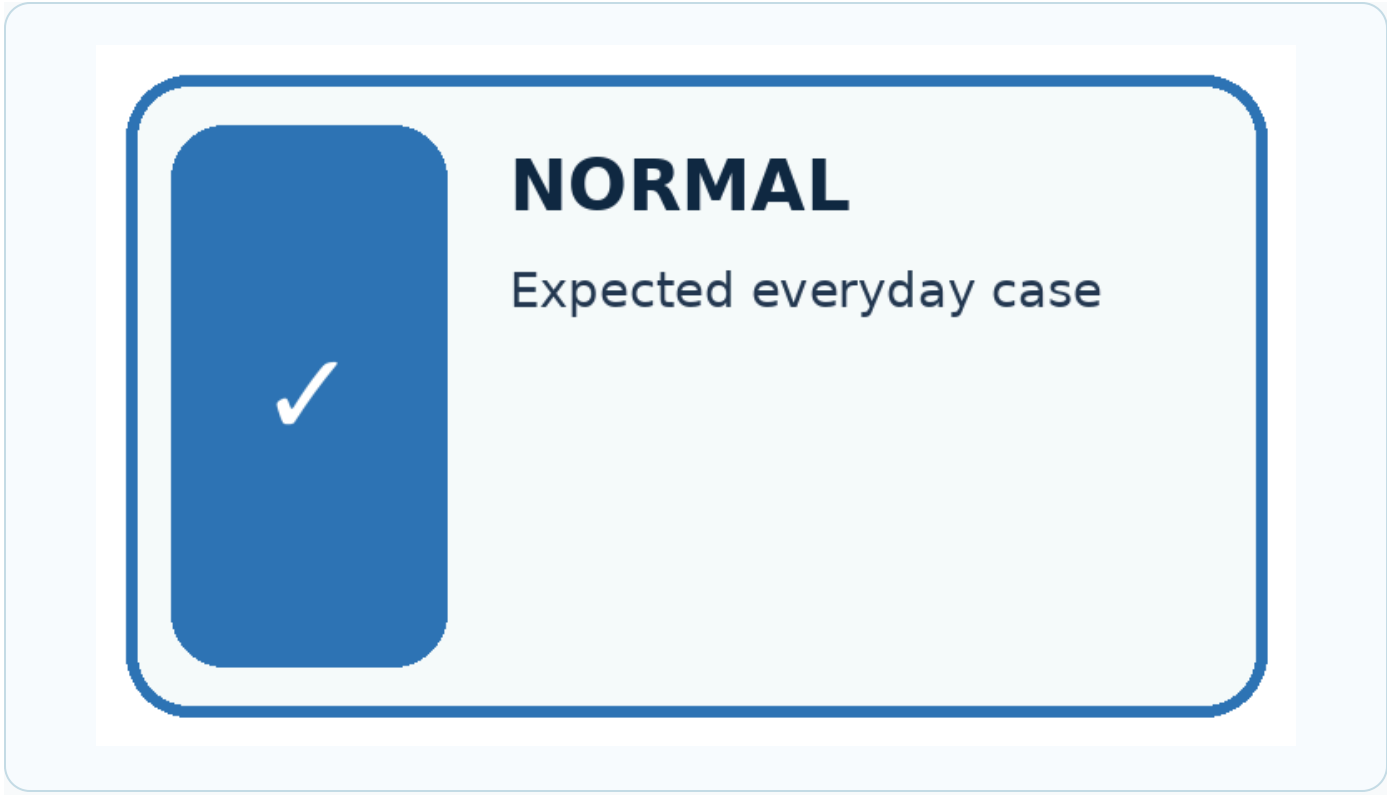
Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

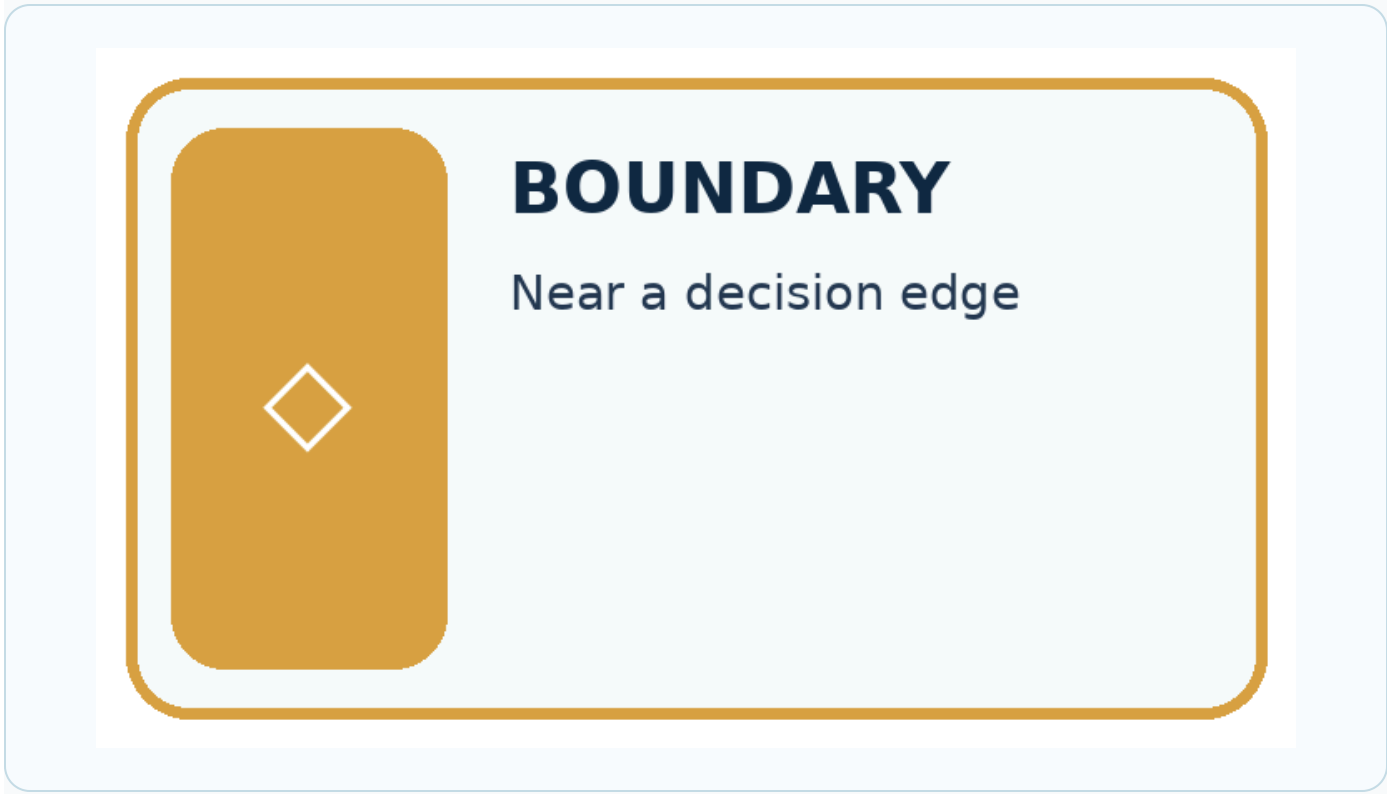
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

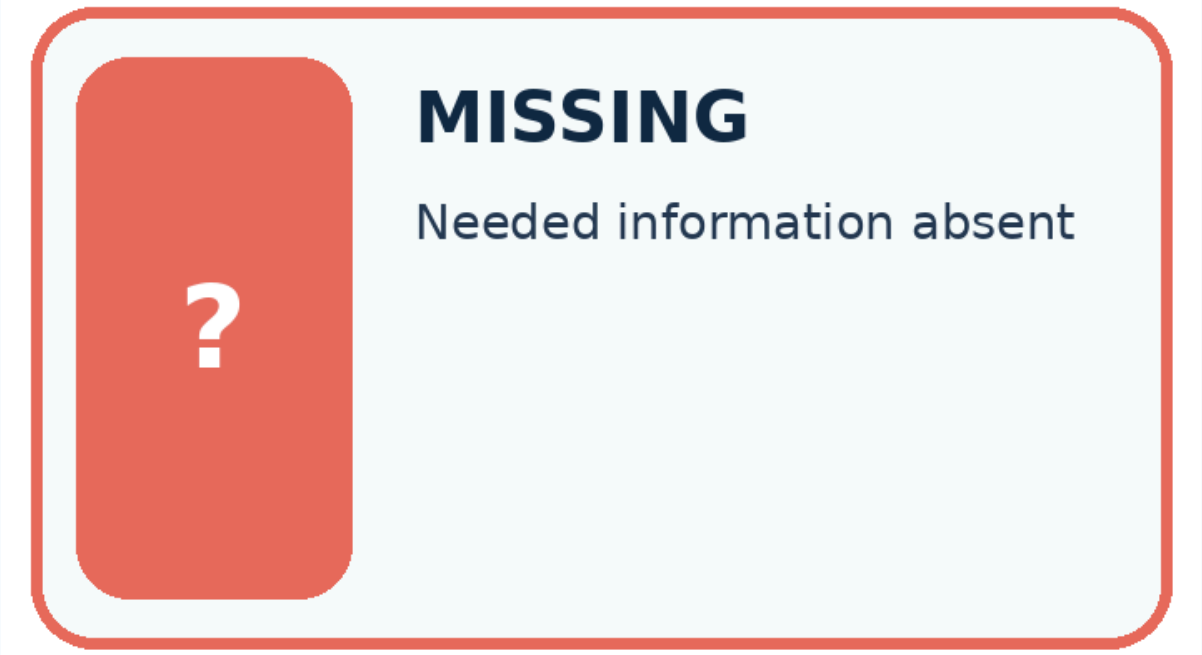
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Explain the centimetre query, then a separate book-format example

> The two training pools have different readings and category names.

Trace the main query's first three neighbours

Three selected neighbours for query 3

STORED EXAMPLE	FEATURE DISTANCE	ONE LABEL VOTE
A at 2 cm	abs 2 minus 3 is 1	blocked
B at 4 cm	abs 4 minus 3 is 1	clear
C at 8 cm	abs 8 minus 3 is 5	clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Store the **training** examples and the method settings. At **prediction** time, compare the query only with stored **training** examples. Do not add the query's reference label or other validation labels to the **neighbour** pool.

No cutoff or tree is fitted here, but retaining **training** examples is still part of the learning method.

Compare k=1 and k=3 before discussing

Why do A and B tie for distance to query 3? Which wins for k=1 under our rule, and why does k=3 predict differently? Does the ID rule establish which label is true?

Keep vote share distinct from a calibrated probability

A **vote** share of two out of three describes these **neighbour**s. It is not a verified two-thirds probability that the new scene is clear. A **prediction** can be wrong even when the nearest examples agree.

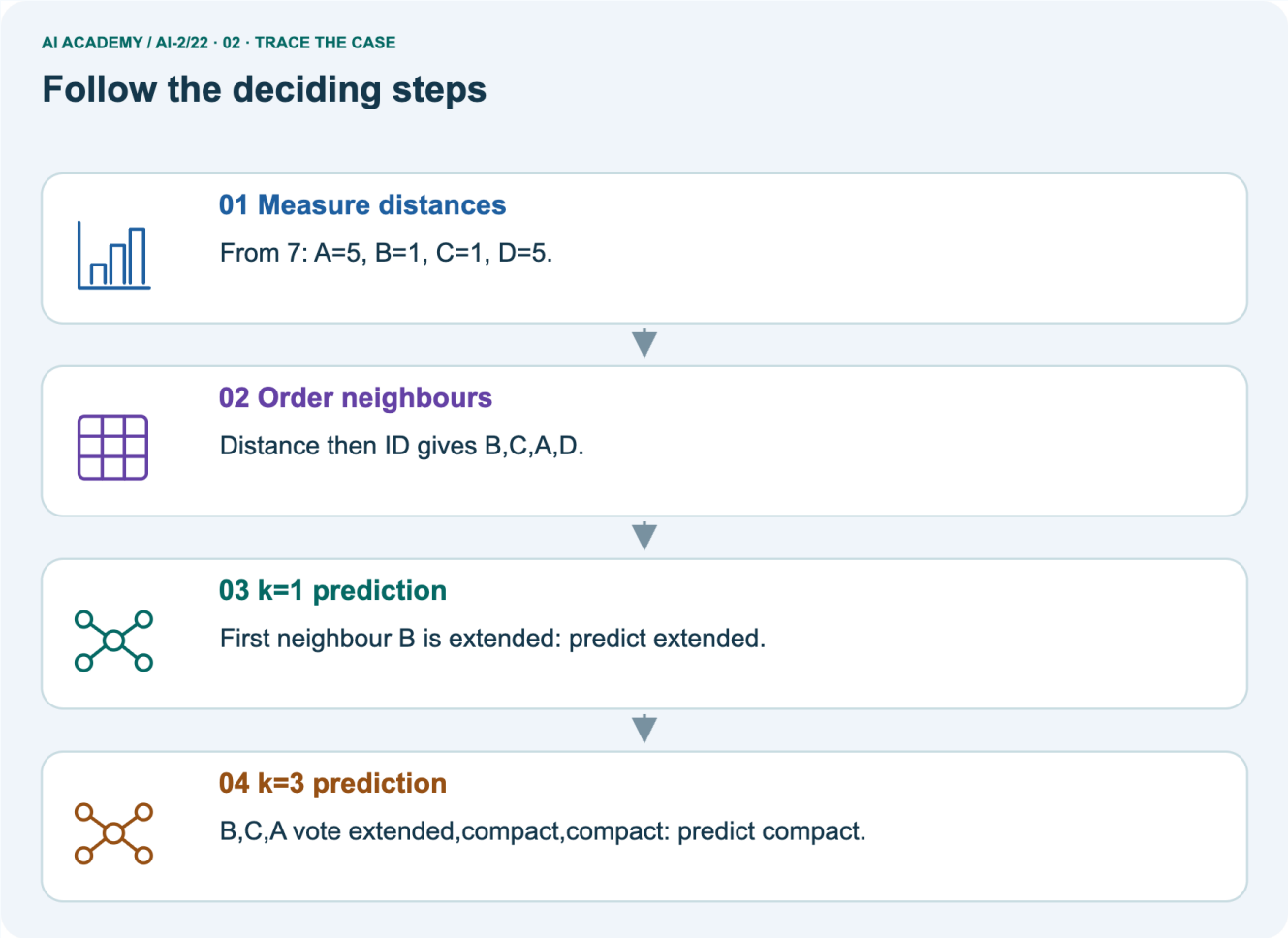
Read the separate four-book pool and tie rules

A fictional one-feature classifier predicts compact or extended book format.

Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

Trace the book-query order and predictions



Follow the labelled picture with the matching explanation. Open the vector diagram

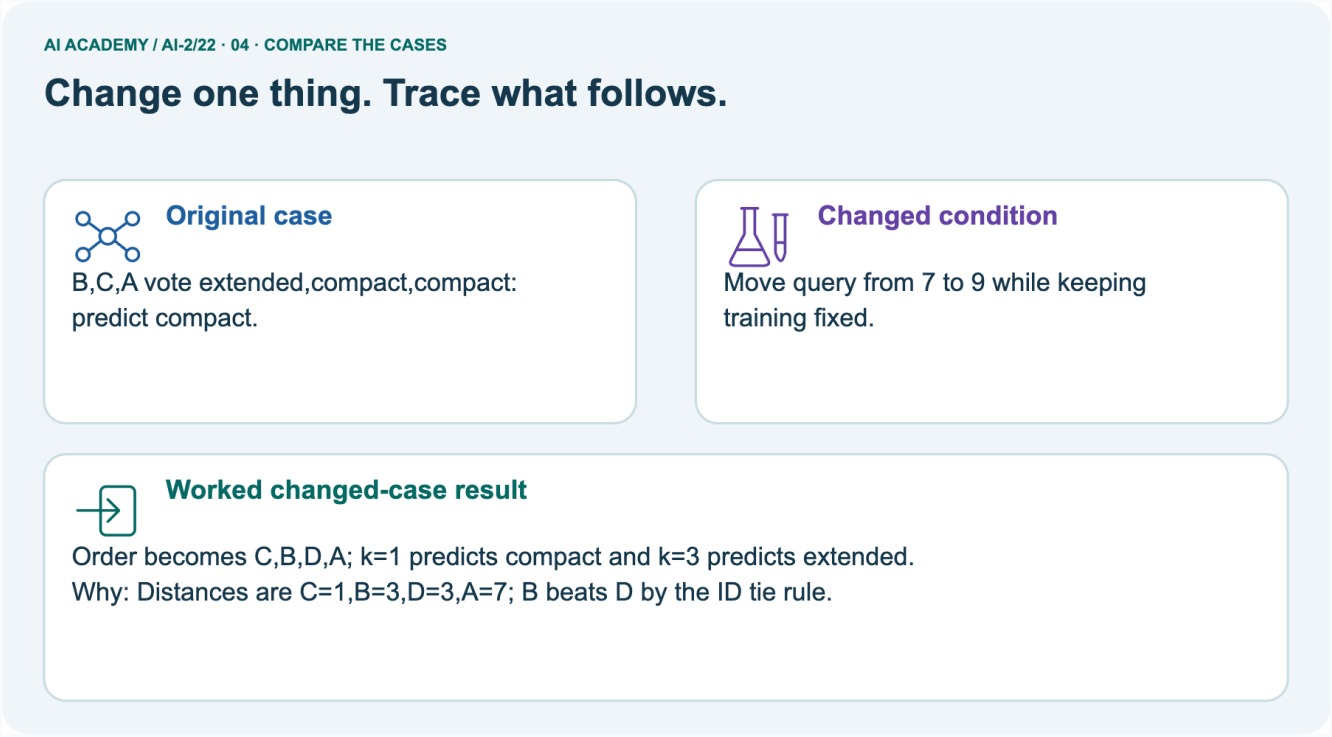
Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Predict the moved book query

- Change one thing: Move query from 7 to 9 while keeping **training** fixed. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain disagreement and the representation limit

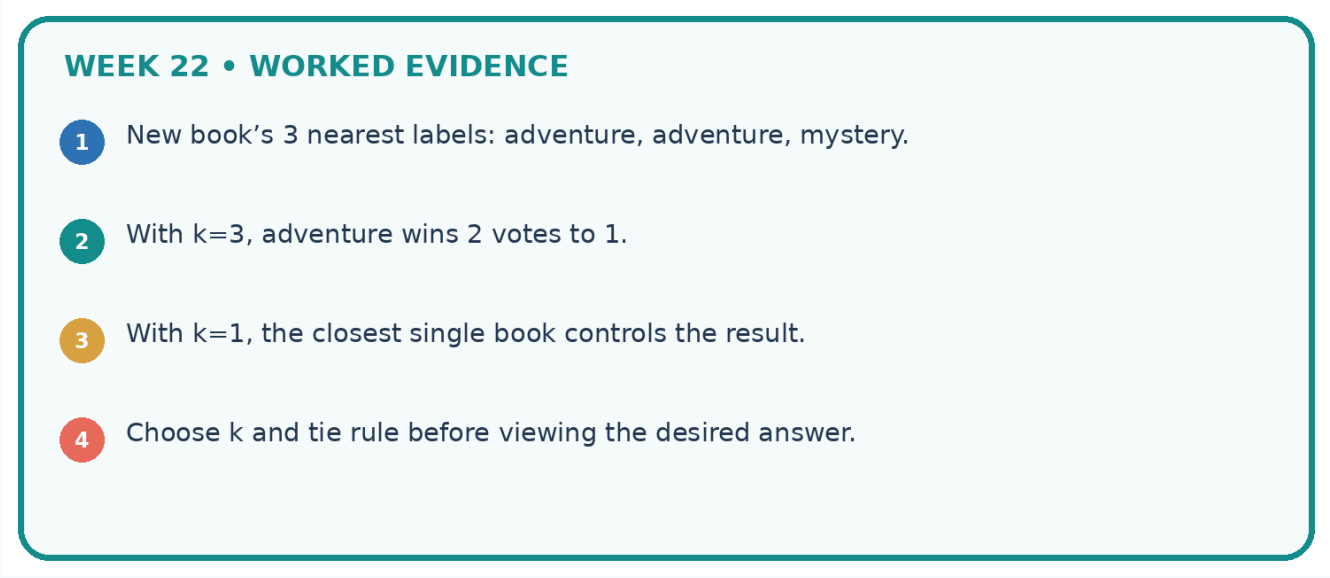
- Read the labelled picture: Why do k=1 and k=3 disagree for the original query?
- Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. New book’s 3 nearest labels: adventure, adventure, mystery.

2. With k=3, adventure wins 2 **vote**s to 1.

3. With k=1, the closest single book controls the result.

4. Choose k and **tie rule** before viewing the desired answer.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

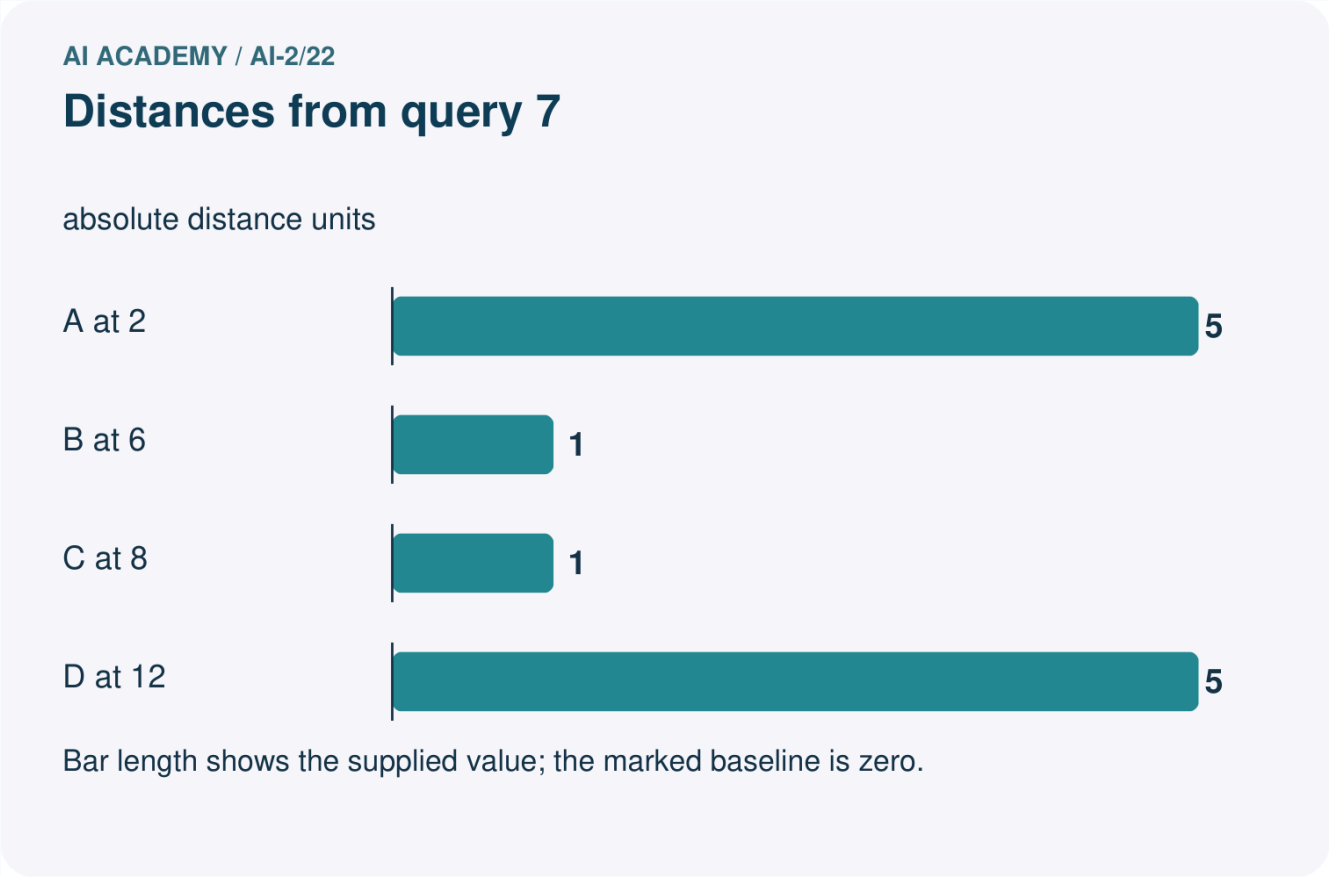
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Nearest Neighbor Reasoning

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/22

Nearest Neighbor Reasoning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

From 7: A=5, B=1, C=1, D=5.



2. Order neighbours

Distance then ID gives B,C,A,D.



3. k=1 prediction

First neighbour B is extended: predict extended.



4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Measure distances	From 7: A=5, B=1, C=1, D=5.
2. Order neighbours	Distance then ID gives B,C,A,D.
3. k=1 prediction	First neighbour B is extended: predict extended.
4. k=3 prediction	B,C,A vote extended,compact,compact: predict compact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

A query must not bring its answer into the training pool

> An exact feature match does not certify a true label.

Retain feature-scale and split limits

All **distance**s here use one feature measured in the same unit. With multiple features, their units and scales affect which examples appear close. A large numerical range can dominate a **distance** calculation. Any learned scaling values must come from **training** only. Next is Week23, Recommendation Scores; retain explicit feature and weighting choices rather than treating raw closeness as objective quality.

Identical readings from different trips can have different labels, and an exact match does not certify truth. Keep related examples grouped across splits as in Week 11. Testing an example against a pool containing itself can make a one-**neighbour** score misleadingly high; it does not show performance on a new trip.

Challenge a self-neighbour perfect score

> **A learner adds each labelled validation query to the stored pool, then uses k=1 and reports perfect predictions. Explain what the result fails to test and how to restore the intended comparison.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The nearest example is always enough for a reliable decision. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Score the frozen method on all three supplied validation queries

> Keep the baseline on the same references.

Read the query references and selected IDs

Keep k=3 and the original five **training** rows. Validation queries with reference labels are (3,blocked), (9,blocked), (11,clear). Their selected **neighbour** IDs are A B C, C D E, and D E C respectively. Each group **vote**s clear.

Only the last **prediction** matches, so accuracy is 1/3, about 33.3 percent.

Compare the training-majority baseline and preserve the final test

The **training**-majority baseline also predicts clear because three **training** labels are clear and two are blocked. It gets the same 1/3 on these validation rows. Record all **distance**s, selected IDs and errors.

Trying more k values is a development choice to document; preserve a separate final test for a final evaluation.

Practise the neighbour/tie rules in W2

Use workbook W2 for the connected example “Nearest **neighbour**s and **distance**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the negative-distance misconception

- Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate book k disagreement

Why do k=1 and k=3 disagree for the original query?

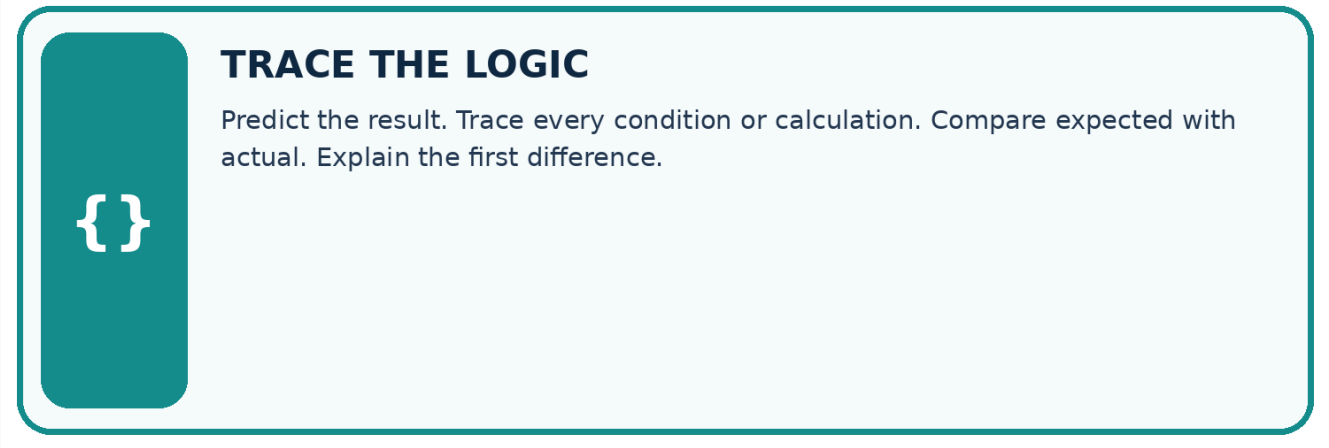
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Practise with less help: Apply neighbour and tie rules

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit the complete stored-pool prediction and validation process

> Do not invent training examples or add validation answers to the pool.

The runnable unchanged-training comparison

```
train = [("A", 2, "blocked"), ("B", 4, "clear"),
         ("C", 8, "clear"), ("D", 10, "blocked"),
         ("E", 12, "clear")]
def predict(query, k):
    ranked = sorted((abs(x - query), item_id, label)
                    for item_id, x, label in train)
    near = ranked[:k]
    labels = [row[2] for row in near]
    prediction = "clear"
    if labels.count("blocked") > labels.count("clear"):
        prediction = "blocked"
    return [row[1] for row in near], prediction
for k in [1, 3]:
    print("query3", k, predict(3, k))
validation = [(3, "blocked"), (9, "blocked"), (11, "clear")]
training_labels = [row[2] for row in train]
baseline = "clear"
if training_labels.count("blocked") > training_labels.count("clear"):
    baseline = "blocked"
correct = baseline_correct = 0
for query, reference in validation:
    ids, prediction = predict(query, 3)
    print(query, ids, prediction, reference)
    correct += prediction == reference
    baseline_correct += baseline == reference
print("validation", correct, len(validation))
print("baseline", baseline, baseline_correct, len(validation))
```

Predict, order, vote, score and preserve

1. Use only the supplied five training rows. Predict all five distances for query3, then selected IDs and labels under k1 and k3. Keep the distance/ID rule unchanged. Next predict all selected IDs for validation queries3,9,11 using k3, before comparing with their supplied references. Run the complete program or label a paper trace. The function receives query and k, not the query reference; the reference is used only after prediction for scoring. Its declared contract requires unique IDs, numeric readings, allowed clear/blocked labels and1<=k<=training count; this teaching code does not implement those validations. Retain each error, selected IDs, settings and training-majority baseline result. Record any later k search as development and preserve a final test. Reserve the new P–T pool for independent W4.

Keep the evidence and scope

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

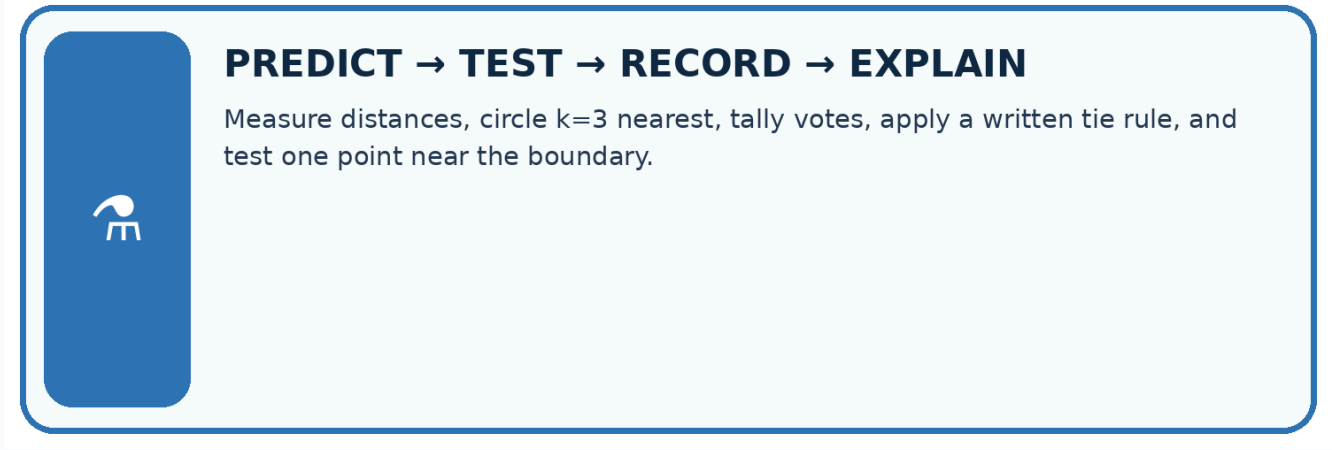
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied five-row training pool, query3 and the three validation records. Record **distance**s, selected IDs, k, **vote**s and predefined ties; compare the frozen k3 method and training-majority baseline without adding query answers to training.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: use the new P–T training pool

> Keep its query and validation records separate from the worked pool.

The independent training and validation brief

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

Method and tie policy

Use numeric absolute distance in centimetres, unique training IDs, distance then alphabetical-ID ordering, and unweighted selected-label votes. Predict before comparing with validation references. The stated binary k1/k3 votes cannot tie; distance ties still require the ID policy. Choose the baseline from the new training labels, not validation answers.

Your first attempt and support record

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Find neighbours independently

Week 22 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute **distance** to choose its nearest **neighbour**.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Use **neighbor**ing liked books as **evidence**, while explaining which traits created closeness.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep neighbour evidence beside the recommender result

> Explain why those examples were close and which information was unavailable.

This week’s concrete contribution

For the fictional recommender, identify the stored training examples, selected features, distance, k and tie policies before explaining a neighbour-based prediction. Keep the new case’s label out of the stored pool. Retain neighbour IDs and distances so a human can inspect the evidence. Liked neighbours and vote fractions do not guarantee a good recommendation or a probability of enjoyment; validate any hard requirements separately.

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

Keep the checkpoint and human decision

THIS WEEK’S CHECKPOINT Use **neighbor**ing liked books as **evidence**, while explaining which traits created closeness.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the stored evidence and the prediction limit

> State distance, neighbour IDs, k, ties and the reference-free prediction path.

Explain without notes

- Close your notes. Explain this goal in your own words: Compute absolute **distance**s and rank **training neighbour**s

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
neighbor	An example close to another under a specified distance measure.	
k	In k-nearest neighbours, the number of nearby examples considered in the prediction.	
vote	A contribution to a choice, such as one neighbour's label in a classification decision.	
tie rule	A predefined method for handling candidates with equal scores or conflicting nearest labels.	

Terms for the connected explanation

Word	Meaning
Distance	A numerical measure of difference between selected feature values

Word	Meaning
Neighbour	A stored training example close under the chosen distance

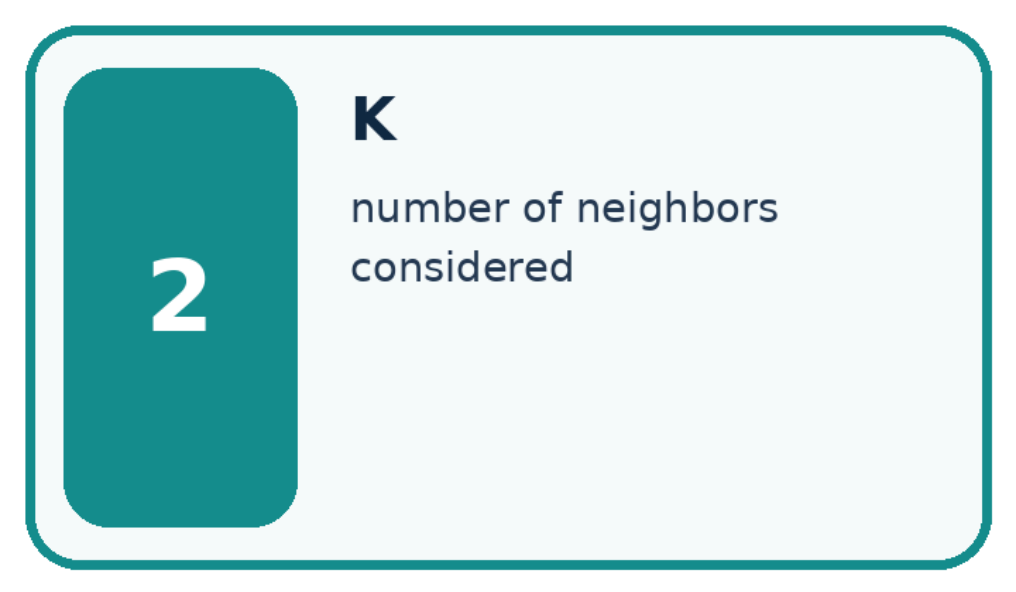
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

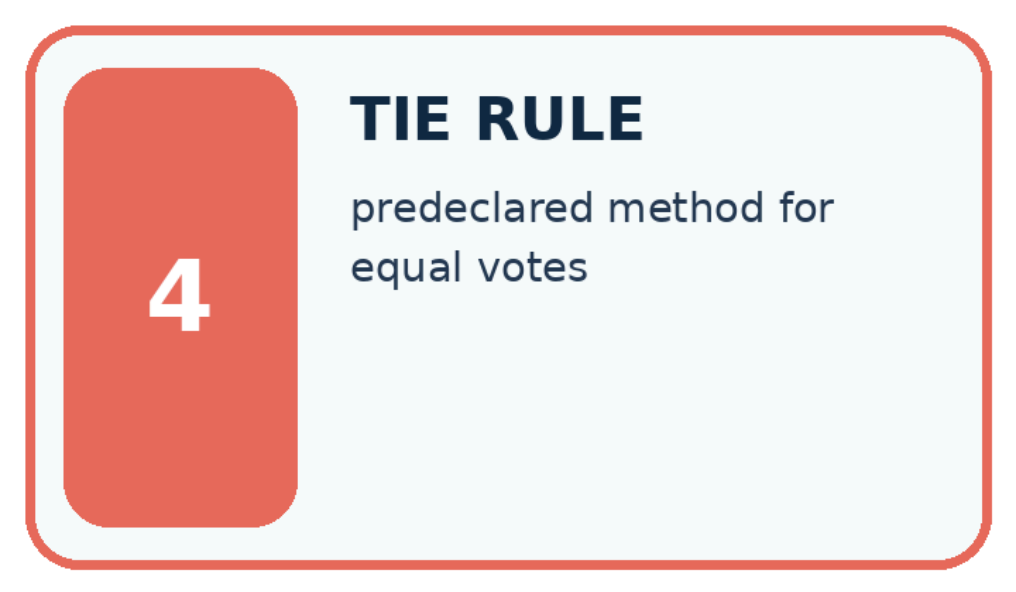
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
K
number of neighbors considered


3
VOTE
combine neighbor labels



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

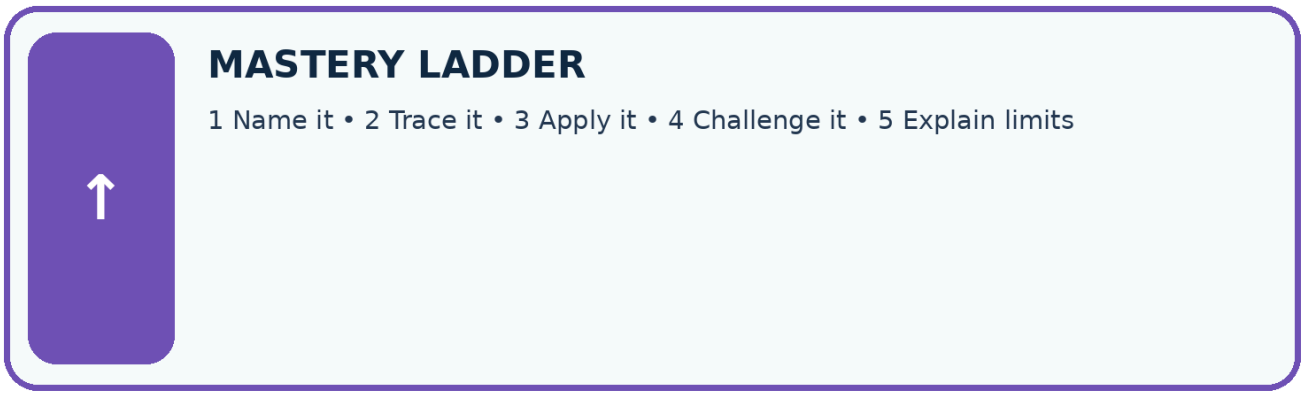
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Find neighbours independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Find neighbours independently is the selected independent case.



EXPLANATION CHECK Keep training storage, numeric distance, k, tie policies and reference-free prediction explicit before evaluation.

Write or draw your reasoning and evidence.

▼ **Extra practice: A nearest-neighbour tie needs a policy**

REFERENCE · SECTION 13

Extra practice: A nearest-neighbour tie needs a policy

> Try the question before opening its worked explanation.

Read the supplied case and question

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest **neighbour**s using differing-trait **distance** and k=1.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Evidence or stage	Value or result
A [1,1], liked	1
B [1,0], disliked	2
C [0,0], disliked	1

▼ Earlier wording, alternatives and picture transcripts

Week 22 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A nearest-neighbour tie needs a policy

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest **neighbour**s using differing-trait **distance** and k=1.

A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Compute absolute distances and rank training neighbours	Explain the deciding step, show a trace or test, and state one limit.
Apply a declared neighbour count and distance tie rule	Explain the deciding step, show a trace or test, and state one limit.
Evaluate neighbour predictions without using validation labels as neighbours	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional one-feature classifier predicts compact or extended book format.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional one-feature classifier predicts compact or extended book format.
- Which labelled step matches this detail? B,C,A **vote** extended,compact,compact: predict compact.
- 1. k=1 **prediction**
- 2. Order **neighbour**s
- 3. k=3 **prediction**

- 4. Measure **distance**s

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: k=3 **prediction** B,C,A **vote** extended,compact,compact: predict compact. Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed.

The query's reference label must not enter the stored pool before predicting. Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Changed-case question

- Changed-case practice: Move query from 7 to 9 while keeping **training** fixed. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Repair and save your work

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

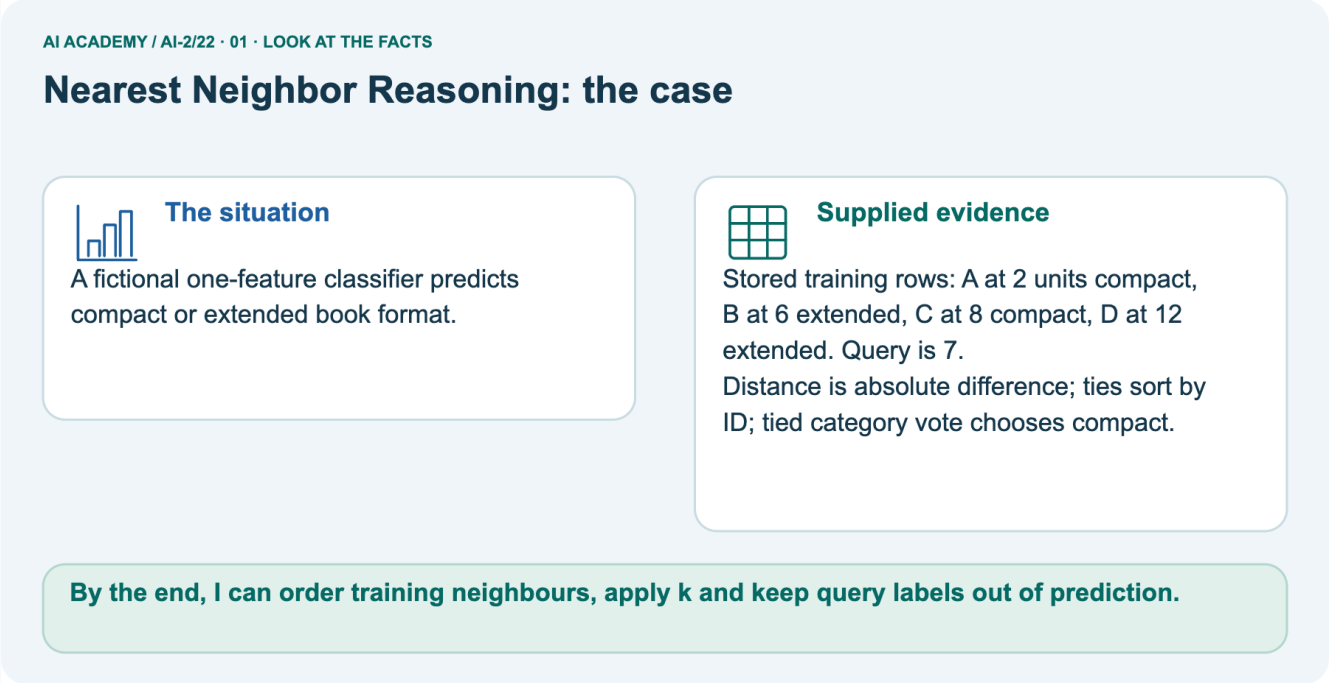
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Nearest Neighbor Reasoning: the case



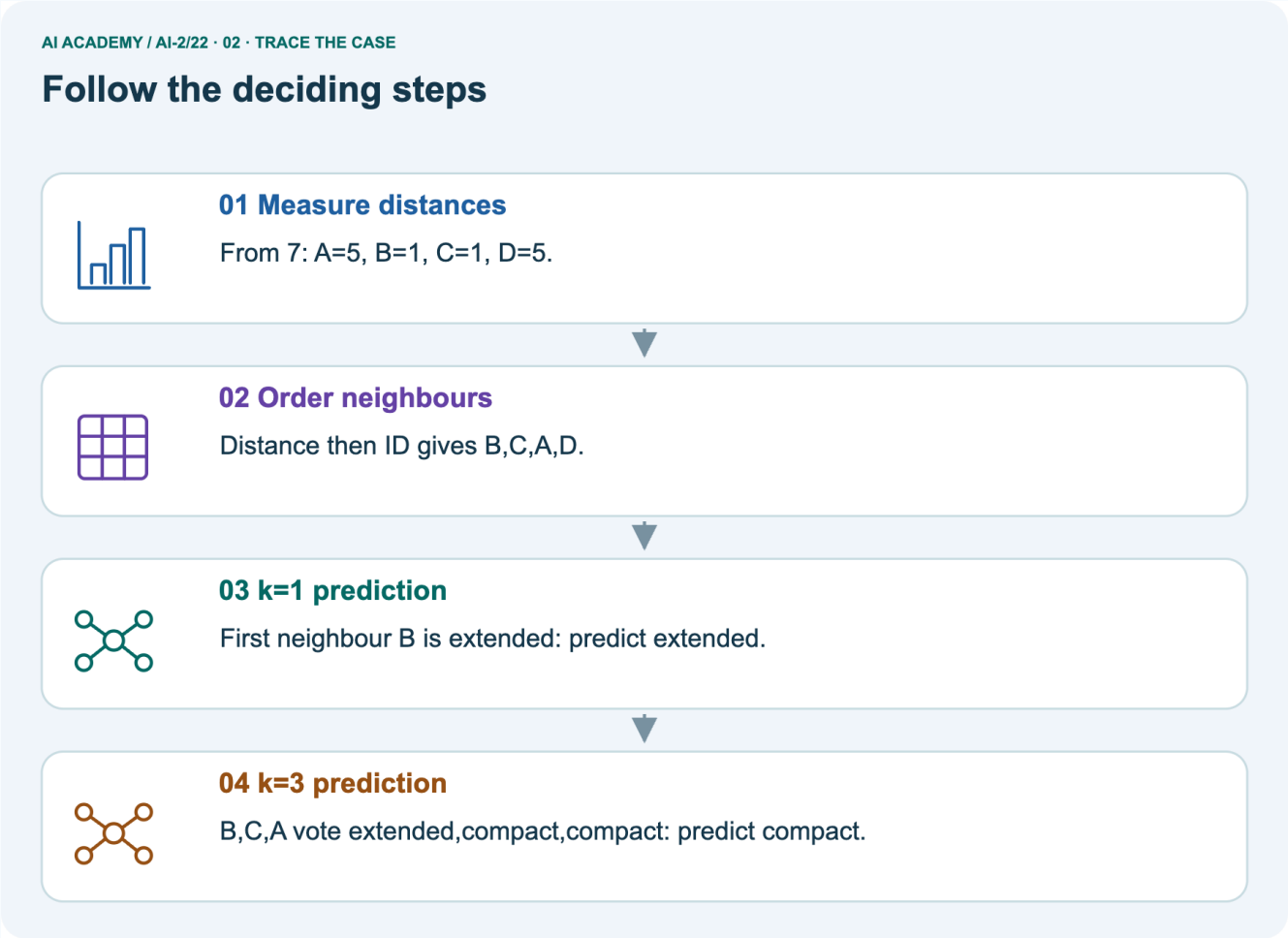
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional one-feature classifier predicts compact or extended book format. Supplied facts: Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

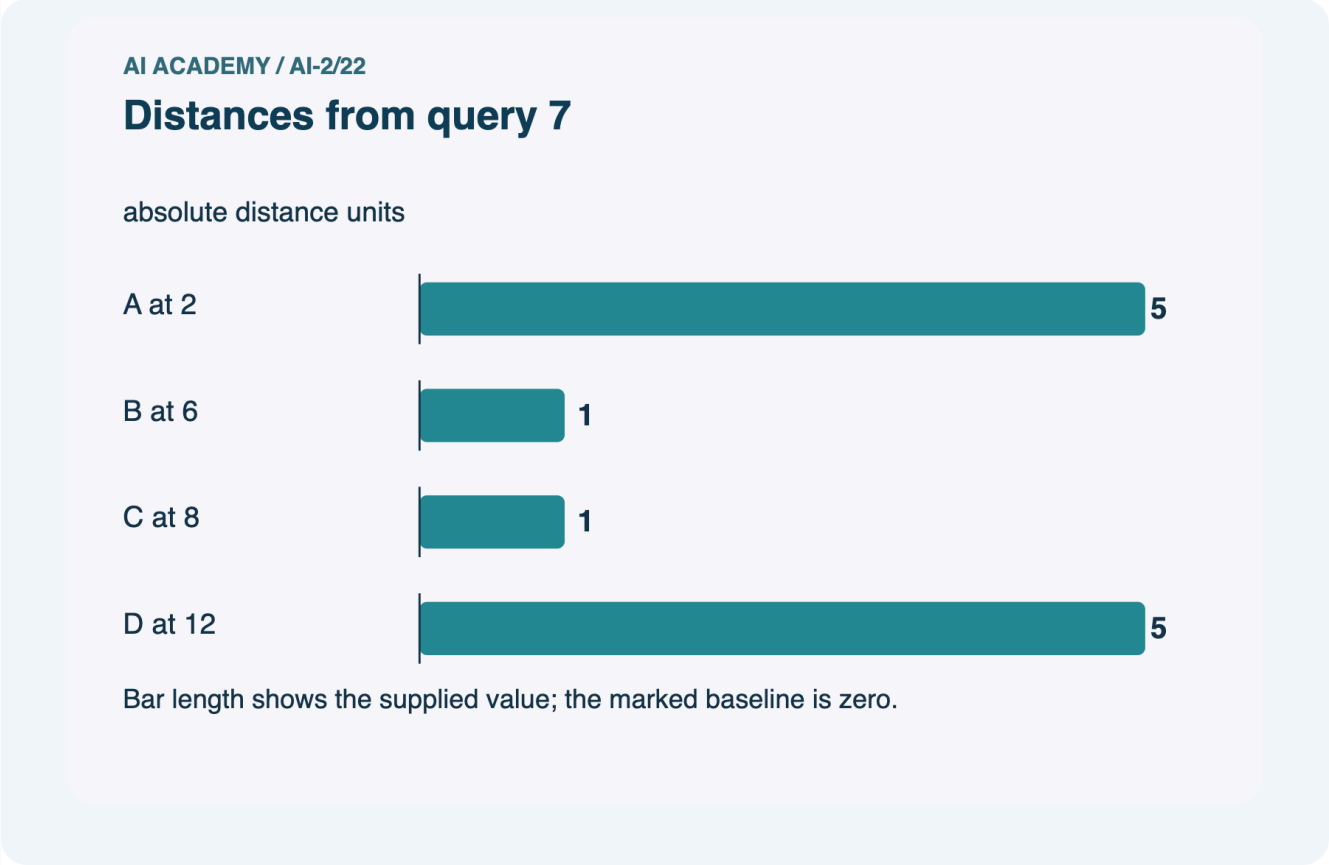


Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: From 7: A=5, B=1, C=1, D=5. → Order **neighbour**s: **Distance** then ID gives B,C,A,D. → k=1 **prediction**: First **neighbour** B is extended: predict extended. → k=3 **prediction**: B,C,A **vote** extended,compact,compact: predict compact.

Point to each numbered stage and explain which recorded **input** it uses.

03. Distances from query 7

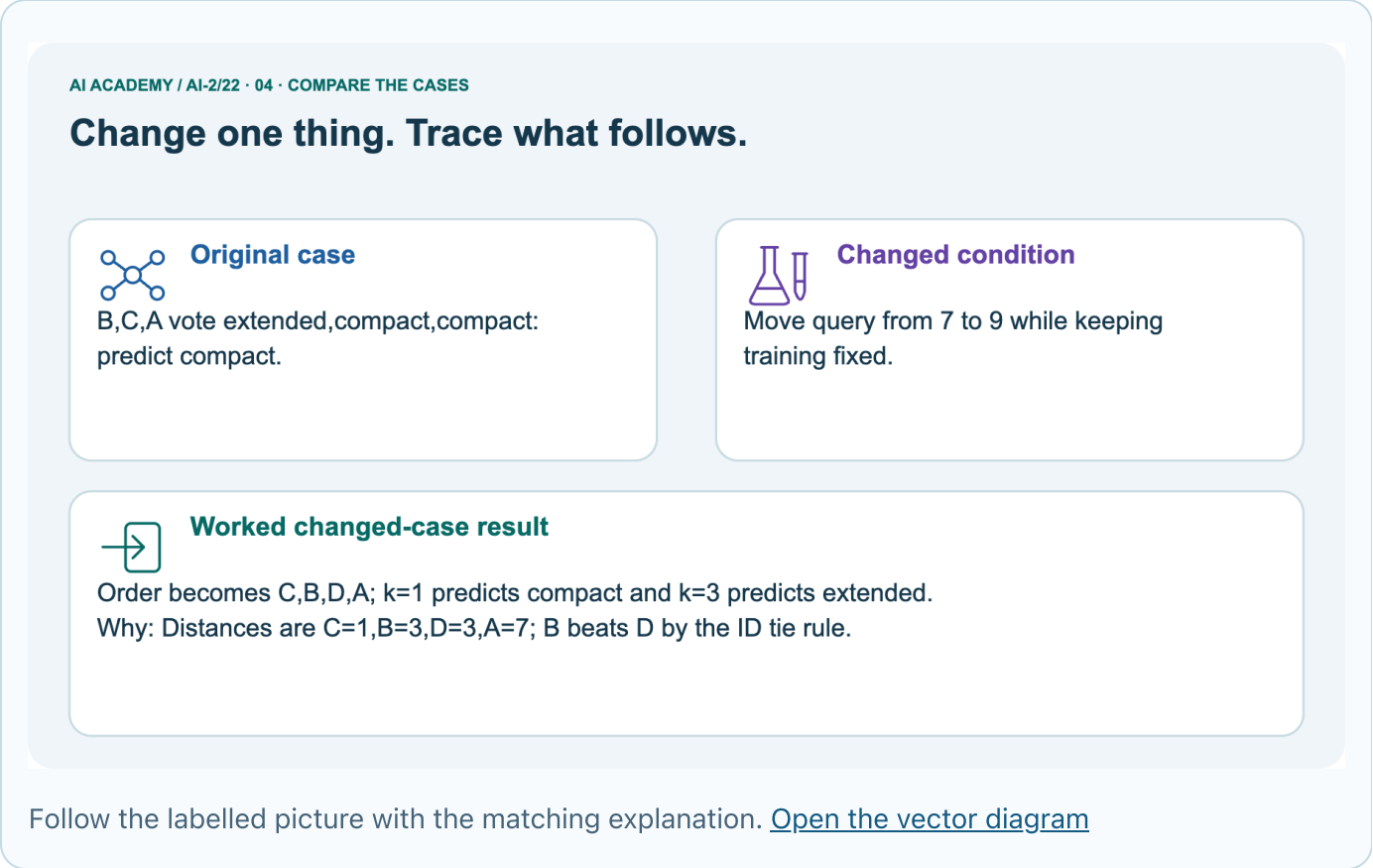


Follow the labelled picture with the matching explanation. Open the vector diagram

Distances from query 7. Units: absolute **distance** units. A at 2: 5. B at 6: 1. C at 8: 1. D at 12: 5.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



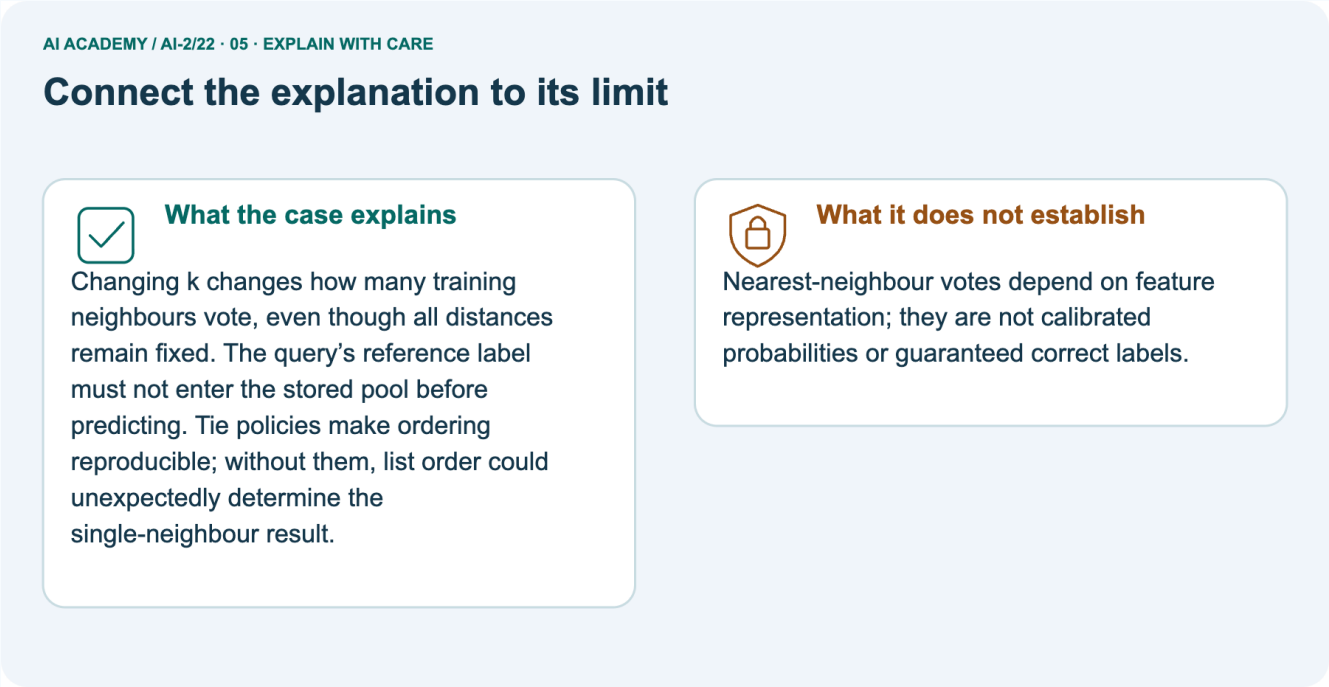
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B,C,A **vote** extended,compact,compact: predict compact. Changed condition: Move query from 7 to 9 while keeping **training** fixed.

Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

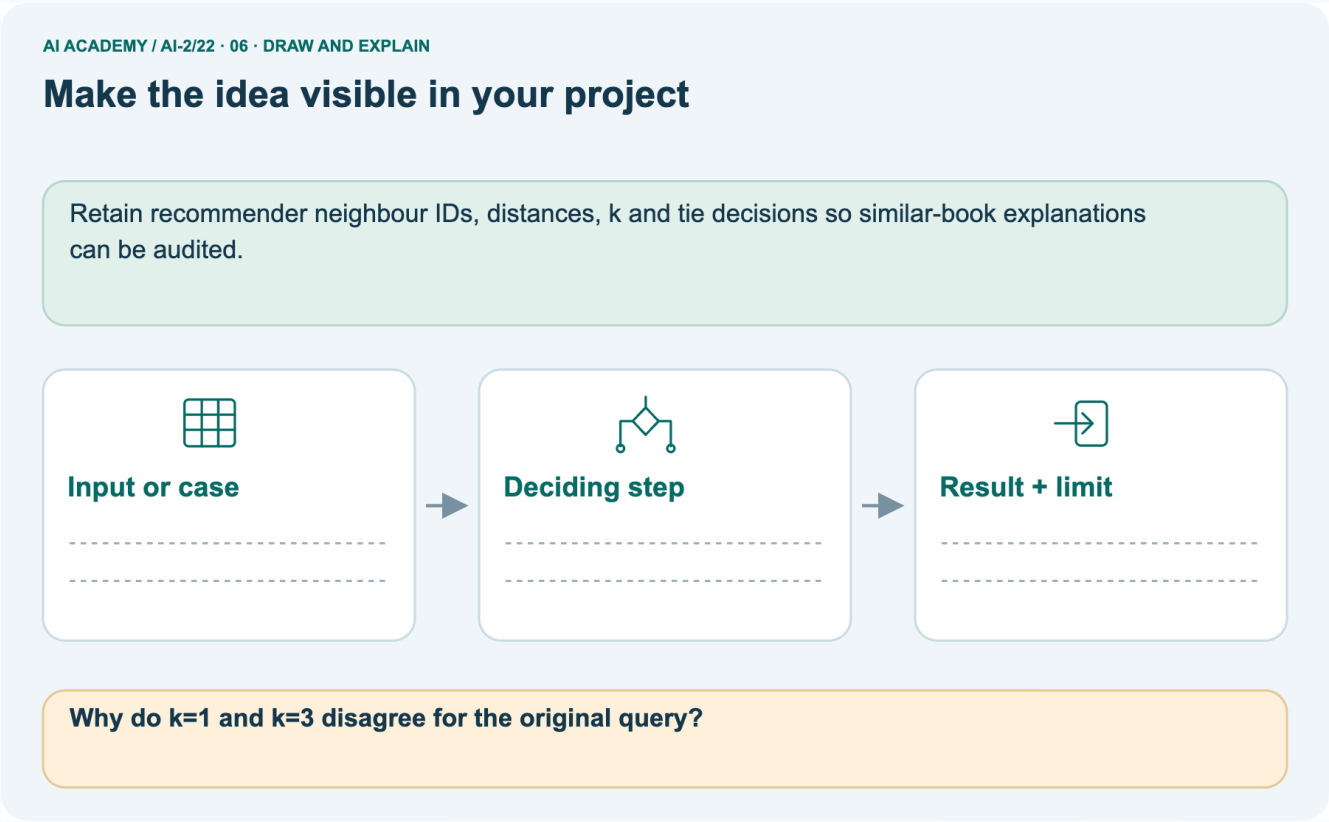
Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Limit: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited. Draw your **input**, deciding step, result and limit.

Supported question: Why do k=1 and k=3 disagree for the original query?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Nearest Neighbor Reasoning: the case

02. Follow the deciding steps

03. Distances from query 7

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.